Lab 5 – Lab Reflection

SRN: PES1UG24AM221

System/Project: Vehicle Service Application – Static Code Analysis

1. What I Learned

In this lab, I learned how static code analysis can be used to identify potential quality issues, style violations, and security-related concerns in Python code without executing the application. I gained experience using analysis tools to assess code quality and improve maintainability.

2. Static Code Analysis Experience

I worked with the Vehicle Service application and used tools including Pylint, Bandit, and Flake8 as part of the analysis workflow. The aim was to review the code for coding-standard issues, suspicious patterns, and potential security problems.

3. Improvements and Verification

I reviewed the reported findings and updated the code and/or checks as needed. The final Pylint result reported a score of 10/10, and Bandit reported zero issues. Flake8 was also part of the verification process; its final result should be recorded only after confirming the completed run.

4. Challenges Faced

A key challenge was understanding which findings represented actual problems and which were style or configuration-related warnings. I also had to run the tools in the available environment and interpret their outputs correctly.

5. Skills Developed

This lab strengthened my skills in Python code review, linting, static security analysis, interpreting tool output, and maintaining readable and consistent code. It also highlighted the value of automated checks during software development.

6. Overall Reflection

Overall, this lab showed how static analysis can catch issues early, before code is run or delivered. Using multiple tools provided different perspectives on code quality and security, and helped me understand how to incorporate automated checks into a development workflow.